

Natural Language Processing

Lecture 3 Bag-of-Words & Vector Space Models

Part 1: Converting Text into
Numerical Representations

Learning Objectives

01 Explain why text must be converted into numbers.

03 Construct a Bag-of-Words representation.

05 Explain Vector Space Models.

07 Measure document similarity using Cosine Similarity.

02 Build a vocabulary from a corpus.

04 Understand Count Vectorization.

06 Calculate TF-IDF values.

Can a Computer Read Text?

HUMANS SEE

"I love Artificial Intelligence."

Meaning is understood immediately through cognitive processing.

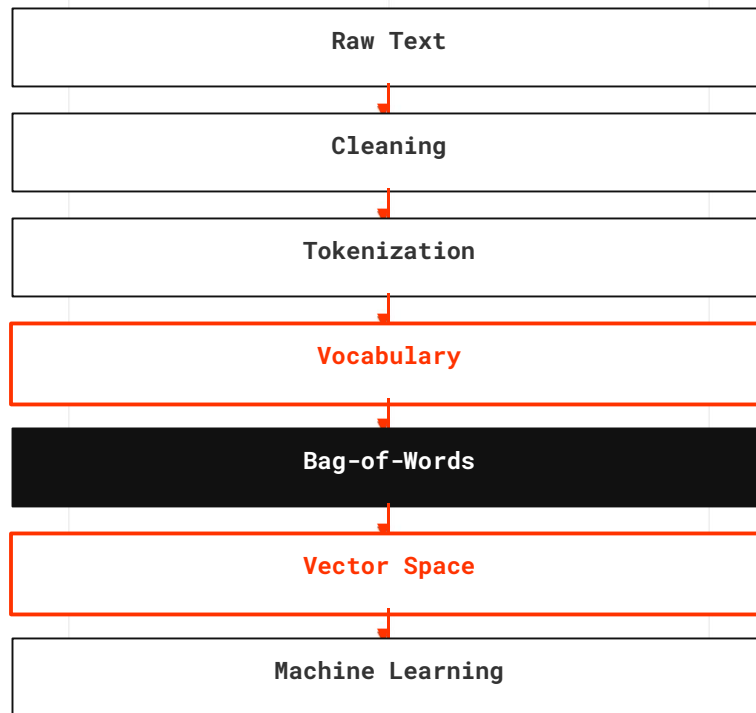
COMPUTERS SEE

["I", "love", "Artificial", "Intelligence"]

↓ Required Format:
[0.24, 0.81, 0.56, 0.92, ...]

THE CHALLENGE: **EVERYTHING** must become **NUMBERS** for a machine to learn.

NLP Representation Pipeline



The Workflow:

Each stage transforms unstructured text into a structured format that algorithms can finally consume.

The Corpus

A **Corpus** is a large and structured collection of machine-readable texts (documents) used for linguistic analysis and model training.

DOCUMENT 1

"I love NLP."

DOCUMENT 2

"NLP is amazing."

DOCUMENT 3

"I love Machine Learning."

Building the Vocabulary

The **Vocabulary** is the set of all **unique words** found across the entire corpus.

Vocabulary Size = 7

Index	Word
1	I
2	love
3	NLP
4	is
5	amazing
6	Machine
7	Learning

Tokenization

The process of breaking down text into smaller units called **tokens**.

"Natural Language Processing is fun."

↓ TOKENS

Natural

Language

Processing

is

fun

PYTHON IMPLEMENTATION

```
text = "Natural Language Processing is fun."  
tokens = text.split()
```

Stop Words Removal

Common words that carry little semantic meaning and are often removed to focus on important terms.

the

is

am

was

to

of

in

and

ORIGINAL

"The students are learning NLP."

PROCESSED

"students", "learning", "NLP"

Stemming: Root Reduction

Playing

Played

Player

Plays



Pla

Stemming uses heuristic rules to chop off word endings to find the base form.

Common Algorithms

Porter Stemmer

Snowball Stemmer

Lancaster Stemmer

Lemmatization

Unlike stemming, **Lemmatization** returns meaningful words (lemmas) by considering the linguistic context.

Provides better linguistic accuracy.

Dictionary-based approach.

Uses morphological analysis of words.

Handles irregular forms (e.g., "better" → "good").

TRANSFORMATION

~~running~~

▼

run

The Bag-of-Words Idea



Ignores Grammar



Ignores Word Order



Only Counts Frequency

CONCEPTUAL TRANSFORMATION

"I love NLP"



{ I: 1, love: 1, NLP: 1 }

Building a BoW Vector

INPUT SENTENCE

"I Love NLP"

I	Love	NLP	Machine	Learning
1	1	1	0	0

[1, 1, 1, 0, 0]

Each position in the vector corresponds to the frequency of a specific vocabulary word.

Natural Language Processing

Part 2: Matrix Construction,
Vector Spaces, & Scikit-Learn

Worked Example: Corpus to Matrix

DOCUMENT 1 (D1)

"I love NLP"

DOCUMENT 2 (D2)

"I love AI"

DOCUMENT 3 (D3)

"AI loves Python"

Resulting Vocabulary

AI

I

Love

Loves

NLP

Python

The Bag-of-Words Matrix

Document	AI	I	Love	Loves	NLP	Python
D1: I love NLP	0	1	1	0	1	0
D2: I love AI	1	1	1	0	0	0
D3: AI loves Python	1	0	0	1	0	1

Each row represents a document vector in the 6-dimensional vocabulary space.

Advantages of Bag-of-Words

01

SIMPLE & INTUITIVE

Easy to understand, explain, and conceptualize for basic NLP tasks.

02

COMPUTATIONALLY FAST

Building the matrix is highly efficient, even for relatively large corpora.

03

EASY IMPLEMENTATION

Requires minimal preprocessing and is supported by all major ML libraries.

04

MODEL COMPATIBILITY

Works exceptionally well with classical models like Naïve Bayes.

05

INTERPRETABLE

Each dimension in the vector corresponds directly to a specific word in the vocabulary.

Limitations of Bag-of-Words

No Word Order

Completely ignores the sequence of words, losing structural meaning.

No Semantics

Cannot distinguish between synonyms or capture context.

Sparsity

Vectors are mostly zeros, leading to inefficient computation.

Dimensionality

Vocabulary size determines vector length, causing "curse of dimensionality".

THE ORDER PROBLEM

"Dog bites man" VS "Man bites dog"

Both produce the exact same BoW vector despite opposite meanings.

Count Vectorizer in Scikit-learn

AUTOMATED WORKFLOW

Creates the vocabulary automatically.

Counts word frequencies per document.

Produces the final sparse BoW matrix.

Handles stop words and tokenization.

PYTHON IMPLEMENTATION

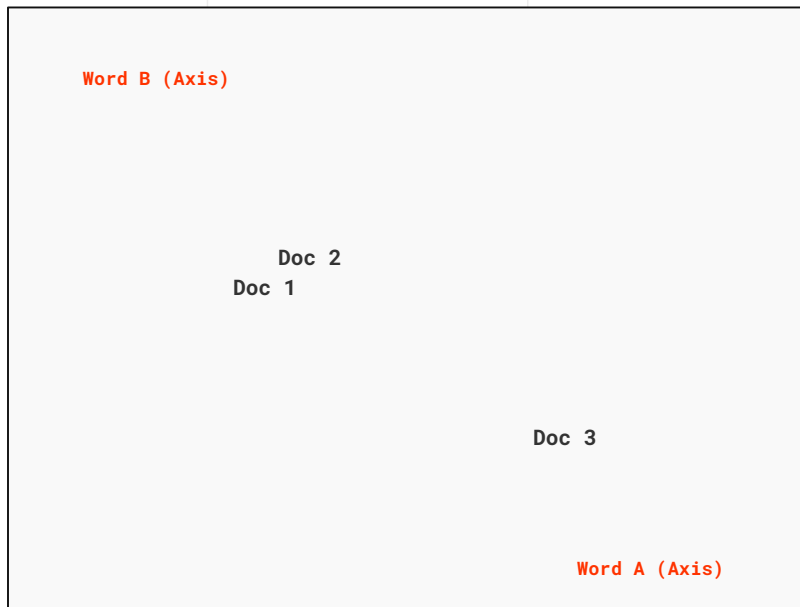
```
from sklearn.feature_extraction.text import CountVectorizer

# Initialize the vectorizer
vectorizer = CountVectorizer()

# Transform the corpus
X = vectorizer.fit_transform(corpus)

# Access the matrix
print(X.toarray())
```

The Vector Space Model (VSM)



Core Concepts

- » **Dimensions:** Each unique word in the vocabulary acts as a geometric axis.
- » **Distance:** Documents with similar word distributions are located closer together.
- » **Meaning:** Proximity in the vector space implies similarity in semantic meaning.

Document Similarity Measures

EUCLIDEAN DISTANCE

Calculates the straight-line distance between two points in the vector space.

MANHATTAN DISTANCE

Calculates the sum of the absolute differences of their coordinates.

COSINE SIMILARITY

Measures the cosine of the angle between two vectors, focusing on orientation rather than magnitude.

MOST POPULAR: Cosine Similarity is the standard choice for NLP tasks due to its robustness against document length.

Cosine Similarity: The Formula

$$\text{Similarity}(A, B) = \frac{(A \cdot B)}{(||A|| * ||B||)}$$

1.0

Identical: Vectors point in the exact same direction (shared vocabulary).

0.0

Orthogonal: Vectors share no common words (completely different).

Example: Cosine Similarity in Action

DOCUMENT 1

"I love NLP"

DOCUMENT 2

"I love AI"

SIMILARITY SCORE

≈ 0.82

High Similarity

Insight: Both documents share the core context ("I", "love"). Even without understanding "NLP" vs "AI", the model recognizes their semantic proximity based on shared word distributions.

Term Frequency (TF)

MATHEMATICAL DEFINITION

$$\text{TF} = \frac{\text{Count of word in doc}}{\text{Total words in doc}}$$

Key Purpose

- » **Normalization:** Prevents bias towards longer documents that naturally have higher word counts.
- » **Local Importance:** Measures how relevant a specific term is to a single document.
- » **Scale:** Typically results in a value between 0 and 1.

Inverse Document Frequency (IDF)

MATHEMATICAL DEFINITION

$$\text{IDF} = \log(N / df)$$

- **N** = Total number of documents
- **df** = Number of documents containing the term

The Core Logic

- » **Rare Words:** Words that appear in few documents are more informative and receive higher weights.
- » **Common Words:** like "the" or "is" appear in almost all documents, resulting in an IDF close to 0.
- » **Normalization:** Helps the model focus on terms that truly distinguish one document from another.

Natural Language Processing

Part 3: TF-IDF Comparison,
Real-World Apps, & Summary

BoW vs. TF-IDF: A Comparison

Feature	Bag-of-Words (BoW)	TF-IDF
Core Logic	Raw Word Counts	Importance Weighting
Considers Importance	No (Equal weighting)	Yes (Rare words prioritized)
Simplicity	Extremely Simple	Slightly more complex
Primary Use Case	Basic classification	Search, Information Retrieval
Model Accuracy	Baseline / Medium	High

Real-World Example: Google Search

01

INDEXING

Google converts millions of webpages into high-dimensional vectors in a massive vector space.

02

QUERYING

Your search query is cleaned and converted into a vector in the same space.

03

MATCHING

The system calculates **Cosine Similarity** between your query and indexed documents.

04

RANKING

Documents with the highest similarity scores are ranked and returned as results.

Summary: What We've Learned

Corpus & Vocabulary

The foundational data structures for all text analysis and model training.

Text Cleaning

Tokenization, Stop Words removal, Stemming, and Lemmatization to normalize input.

Bag-of-Words (BoW)

A simple frequency-based representation that ignores grammar and word order.

Vector Space Model

Conceptualizing documents as points in a high-dimensional geometric space.

Cosine Similarity

The standard metric for measuring semantic closeness between document vectors.

TF-IDF Weighting

Advanced representation that prioritizes rare, informative words over common ones.

Knowledge Check: Quiz

01

What does the Bag-of-Words (BoW) model ignore?

Answer: **Grammar and Word Order**

02

Which technique gives more importance to rare words in a corpus?

Answer: **TF-IDF (Term Frequency-Inverse Document Frequency)**

03

Which similarity measure is most widely used for comparing text vectors?

Answer: **Cosine Similarity**

Next Lecture & Activity

LOOKING AHEAD

Lecture 4: Naïve Bayes & Smoothing

Bayes Theorem & Conditional Probability

Naïve Bayes Classifier for Text

Laplace Smoothing & Spam Detection

CLASSROOM ACTIVITY (15 MINS)

1. Create a corpus of **five sentences**.
2. Build the **vocabulary** manually.
3. Construct the **Bag-of-Words matrix**.
4. Discuss: How would **TF-IDF** change your matrix values?